

# Pedro Masi

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Haslam College of Business, University of Tennessee, Knoxville, TN 37916

U.S Permanent Resident (Green Card Holder)

- [LinkedIn](#)
- [Website](#)

## EDUCATION

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|---------------------------------------------------------------------------------------|----------------------------|
| <b>University of Tennessee, Knoxville</b><br>PhD in Business Analytics and Statistics | <i>July-2023 - Present</i> |
| <b>Vrije Universiteit Amsterdam</b><br>M.S in Finance and Technology                  | <i>Aug-2020 - Dec-2021</i> |
| <b>Kansas State University</b><br>M.S in Agricultural Economics                       | <i>Aug-2017 - May-2019</i> |
| <b>Kansas State University</b><br>BA in Agricultural Economics                        | <i>Aug-2011 - May-2015</i> |

## RESEARCH INTERESTS

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My research lies at the intersection of AI and causal inference. I apply text mining and embedding techniques to measure technology adoption and its impact on labor and financial outcomes. I convert unstructured patent and job posting text into structured representations that feed into econometric models to estimate the effects of AI exposure on skill value and firm capital investments.

**Keywords:** • Machine Learning • Representation Learning • NLP • Causal Inference • Reinforcement Learning

## WORKING PAPERS

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Masi Pedro, Yuanyang Liu, and Chuanren Liu. **“AI Upskilling: Decoupling Skill Demand and Skill Premium”** - under review ISR (Information Systems Research)

Masi Pedro, Howard Zhong, Yuanyang Liu, and Chuanren Liu. **“AI Skill Herding and Capital Investment: When Does Following the Crowd Pay Off?”** In preparation for submission - target (MISQ)

## PAPERS IN PROGRESS

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Masi Pedro, Yuanyang Liu, and Chuanren Liu. **“Decoding AI Innovation: A Transformer-Based Approach to Patent Landscaping”** In preparation for submission

## CONFERENCE PRESENTATIONS AND INVITED TALKS

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- **“AI Skill Herding and Capital Investment: When Does Following the Crowd Pay Off?”**
  1. Accepted (presentation forthcoming), @ Americas’ Conference on Information Systems (AM-CIS), Reno, NV, August 2026
  2. Accepted (presentation forthcoming), @ Computational Methods in Management Research Workshop, University of Notre Dame (AIS SIG TECH), June 2026.

- **“AI and the Market Value of Skills”**— Accepted @ DePaul Doctoral/Junior Faculty Research Symposium, Chicago, IL, March 2026
- **“AI Upskills Human-Centric and Fundamental Computer Science Knowledge”**
  1. Accepted @ Workshop on Information Technologies & Systems (WITS), Nashville, TN, Dec 2025
  2. Accepted @ Conference of Information System & Technology (CIST), Atlanta, GA, Oct 2025
- Presented AI-Website on “Tracking the New Frontiers of Work in Tennessee” at AI Tennessee Workshop, Knoxville, TN, Jan-2025
- Presented “AI Talk: Can Executives keep their promises?” at INFORMS Annual Conference, Seattle, WA, Oct-2024
- Presented research on “AI for Business Innovation” at Business Analytics Forum, University of Tennessee, Aug-2024

## GRANTS

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**AI Tennessee Initiative Seed Funds: Tracking the New Frontiers of Work in Tennessee.** Liu, Chuanren. (Lead PI), Liu, Yuanyang (Co-PI). Total Award: \$64,800, 2024-2025. *Collaborator.* Responsibilities include coordinating research efforts and providing overall management of project activities; leading empirical economic modeling; conducting economic analysis; overseeing data collection and management; presenting research findings and writing peer-reviewed publications; and supervising four M.S. students ([project website](#)).

## TEACHING EXPERIENCE

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### Instructor

- STAT 201 Introduction to Statistics, University of Tennessee *Fall 2025*

### Guest Lectures

- Guest lecture on “Clustering Techniques” for Data Mining (BZAN 542), University of Tennessee, Knoxville, TN — Sept 2025.
- Guest lecture on “Probability Theory” for Introductory Statistics (Stats 201), University of Tennessee, Knoxville TN-Mar 2025

### Teaching Assistant, University of Tennessee

- STAT 505 Quantitative Methods MBA, Prof Allen Pannell *Spring 2026*
- BZAN 557 Text Mining, Prof Wenjun Zhou *Spring 2026*
- STAT 201 Introduction to Statistics, Prof. Adam Spanbauer *Summer 2025*
- BUAD 503 Business Communications MBA, Prof Angel Norman *Summer 2024*
- BZAN 555 Supply Chain Analytics, Prof. Sean Willems *Spring 2024*
- BZAN 542 Data Mining Methods, Prof. ChuanRen Liu *Fall 2023*

## SELECTED COURSEWORK

| Class Title              | Instructor(s)                 | Year |
|--------------------------|-------------------------------|------|
| Machine Learning         | FNU Suya                      | 2025 |
| Deep Learning            | Michel Ballings               | 2025 |
| Reinforcement Learning   | Michel Ballings               | 2025 |
| Microeconomic Theory     | James Lake                    | 2023 |
| Data Science             | Yuanyang Liu                  | 2023 |
| Design of Experiments    | Robert Mee                    | 2022 |
| Time Series Econometrics | Luiz Lima                     | 2022 |
| Econometrics I & II      | Celeste Carruthers, Luiz Lima | 2022 |
| Applied Optimization     | ChuanRen Liu                  | 2022 |

## PROFESSIONAL DEVELOPMENT

**Causal Inference I** focused on IV, regression discontinuity, matching methods  
**Causal Inference II** focused on diff-in-diff methods  
**Causal Inference III** focused on synthetic control methods  
**MLx Fundamentals** focused on supervised ML, clustering, classification, factorization techniques

## TECHNICAL SKILLS

**Research Methods:** TF-IDF, Word2Vec, LLM2Vec, topic modeling, document clustering. EDA, feature engineering, dimensionality reduction, supervised and unsupervised learning, model tuning and cross-validation.  
**Programming:** Python (pandas, numpy, scikit-learn, pytorch, tensorflow, keras), R, Stata, Sql, PySpark, HPC, Databricks  
**Data & Visualization:** PowerBI, Tableau, JMP, WordPress, QlikView.  
**Writing** L<sup>A</sup>T<sub>E</sub>X, HTML.  
**Languages** English (fluent), Spanish (fluent), Portuguese (fluent)

## WHITE PAPERS

**ESG scores & financial performance of Emerging Market Firms**, VU, Amsterdam [link](#)  
**Climate Aligned Investment Portfolios**, VU, Amsterdam [link](#)  
**Latin America State of the Bonds Market Outlook**, Climate Bonds, IDB, IFC [link](#)

## PROFESSIONAL EXPERIENCE

**Agri-Business Analyst** Rabobank, NL *July-2022 - July-2023*  
**Business Analyst** Rabobank, NL *July-2021 - June-2022*  
**Fixed Income Consultant** Climate Bonds, NL *Mar-2021 - June-2021*  
**Financial Analyst Consultant** University of Nebraska, Lincoln *Aug-2019 - May-2020*

## AWARDS AND SCHOLARSHIPS

**Graduate Research Award**, AMCIS 2026 *August-2026*  
**Graduate Research Award**, DePaul 2026 *March-2026*  
**Graduate HCB Award**, WITS 2025 *October-2025*  
**Graduate GSS Award**, CIST 2025 *October-2025*  
**Graduate GSS Award**, INFORMS 2024 *October-2024*

## ACTIVITIES

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### **Volunteer, Robotics Education for Kids - Centro Hispano**

*Spring 2023*

Assisted in teaching basic robotics concepts to elementary school students, including programming and building simple robots, while fostering enthusiasm for STEM subjects through interactive and engaging activities.